Ejercicios muy sencillos con FreeRADIUS

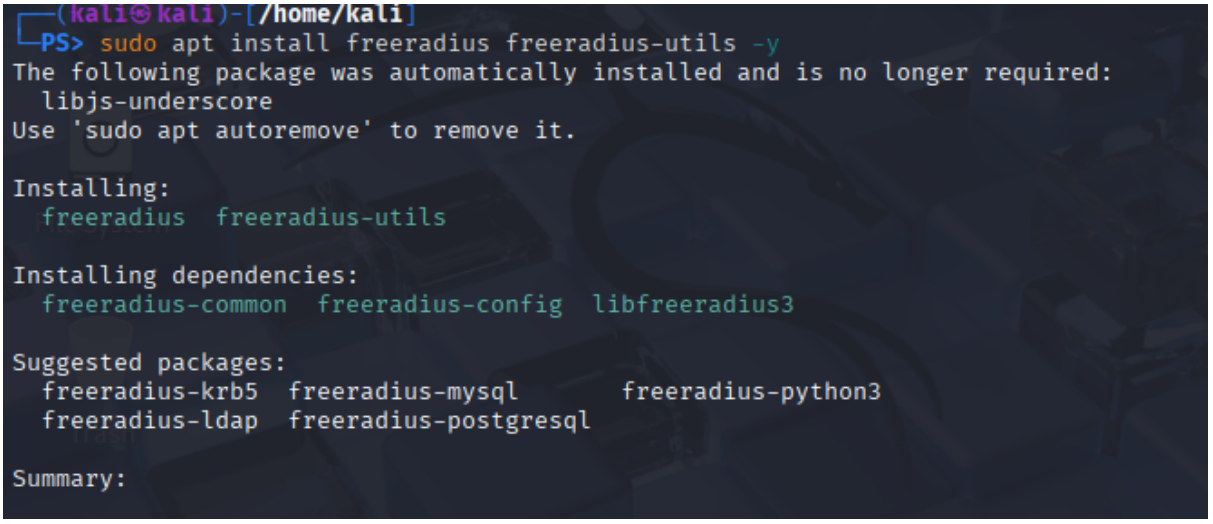
*(Todos pensados para **Kali/Ubuntu en laboratorio**, sin meternos aún en integraciones complejas)*

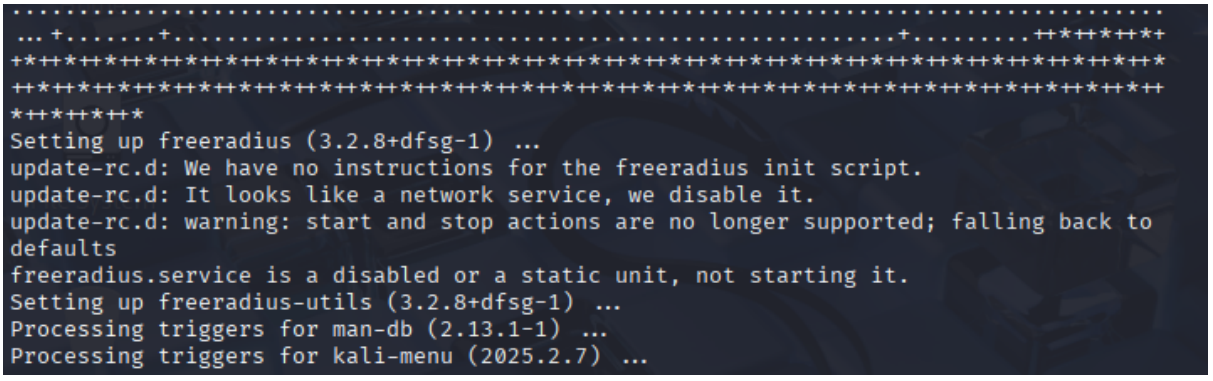
🔹 **Ejercicio 1 — Instalar FreeRADIUS**

sudo apt update

sudo apt install freeradius freeradius-utils -y

👉 Resultado esperado: FreeRADIUS instalado en el sistema.





🔹 **Ejercicio 2 — Comprobar que el servicio corre**

™

```
┌──(kali㉿kali)-[/home/kali]
└─PS> sudo freeradius

┌──(kali㉿kali)-[/home/kali]
└─PS> sudo systemctl status freeradius
○ freeradius.service - FreeRADIUS multi-protocol policy server
     Loaded: loaded (/usr/lib/systemd/system/freeradius.service; disabled; preset: di>
     Active: inactive (dead)
       Docs: man:radiusd(8)
             man:radiusd.conf(5)
             http://wiki.freeradius.org/
             http://networkradius.com/doc/
lines 1-7/7 (END)
```

👉 Resultado esperado: servicio activo (running).

🔹 **Ejercicio 3 — Ejecutar FreeRADIUS en modo debug**

sudo freeradius -X

👉 Resultado esperado: ver en pantalla todos los logs de autenticación (muy útil para aprender).

```
┌──(kali㉿kali)-[/home/kali]
└─PS> sudo freeradius -X
FreeRADIUS Version 3.2.8
Copyright (C) 1999-2025 The FreeRADIUS server project and contributors
There is NO warranty; not even for MERCHANTABILITY or FITNESS FOR A
PARTICULAR PURPOSE
You may redistribute copies of FreeRADIUS under the terms of the
GNU General Public License
For more information about these matters, see the file named COPYRIGHT

FreeRADIUS is developed, maintained, and supported by InkBridge Networks.
For commercial support, please email sales@inkbridgenetworks.com
https://inkbridgenetworks.com/
Starting - reading configuration files ...
including dictionary file /usr/share/freeradius/dictionary
including dictionary file /usr/share/freeradius/dictionary.dhcp
```

```
File  Actions  Edit  View  Help
led/inner-tunnel:366
Compiling Post-Auth-Type REJECT for attr Post-Auth-Type
} # server inner-tunnel
radiusd: #### Opening IP addresses and Ports ####
listen {
        type = "auth"
        ipaddr = *
        port = 0
    limit {
        max_connections = 16
        lifetime = 0
        idle_timeout = 900
    }
Failed binding to auth address * port 1812 bound to server default: Address already in
 use
/etc/freeradius/3.0/sites-enabled/default[59]: Error binding to port for 0.0.0.0 port
1812
```

HAY QUE ABRIR OTRO TERMINAL IMPORTANTE

- **Ejercicio 4 — Añadir un usuario simple**

1. Edita el archivo:

   sudo nano /etc/freeradius/3.0/users

2. Añade al final:

   alumno Cleartext-Password := "1234"

```
File  Actions  Edit  View  Help
  GNU nano 8.4                  /etc/freeradius/3.0/users *
# DEFAULT
#       Service-Type = Administrative-User

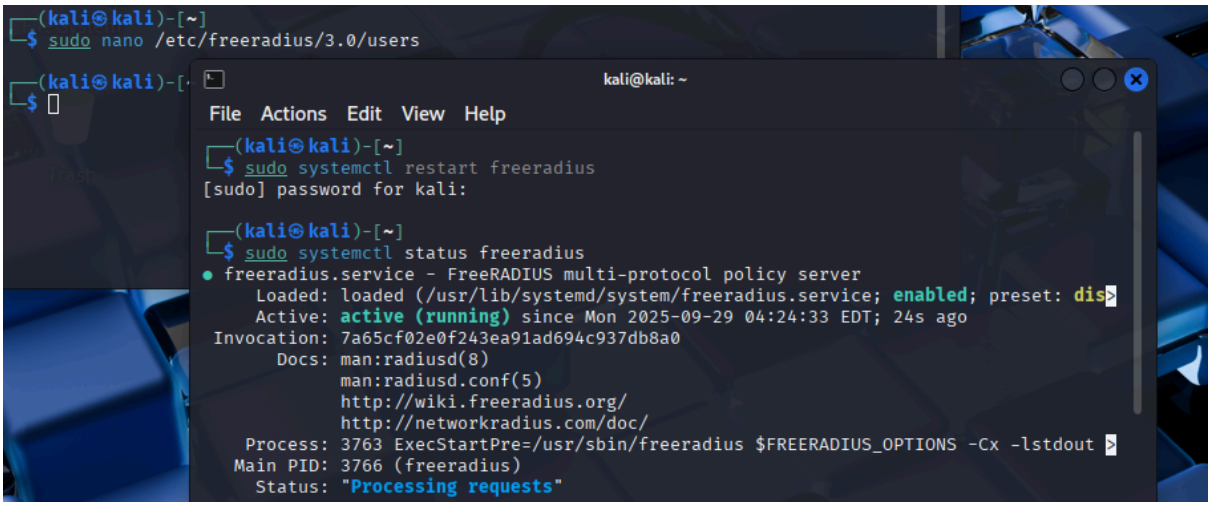
# On no match, the user is denied access.

#############################################################
# You should add test accounts to the TOP of this file! #
# See the example user "bob" above.                     #
#############################################################

alumno Cleartext-Password :="1234"
```

3. Guarda y reinicia:

sudo systemctl restart freeradius

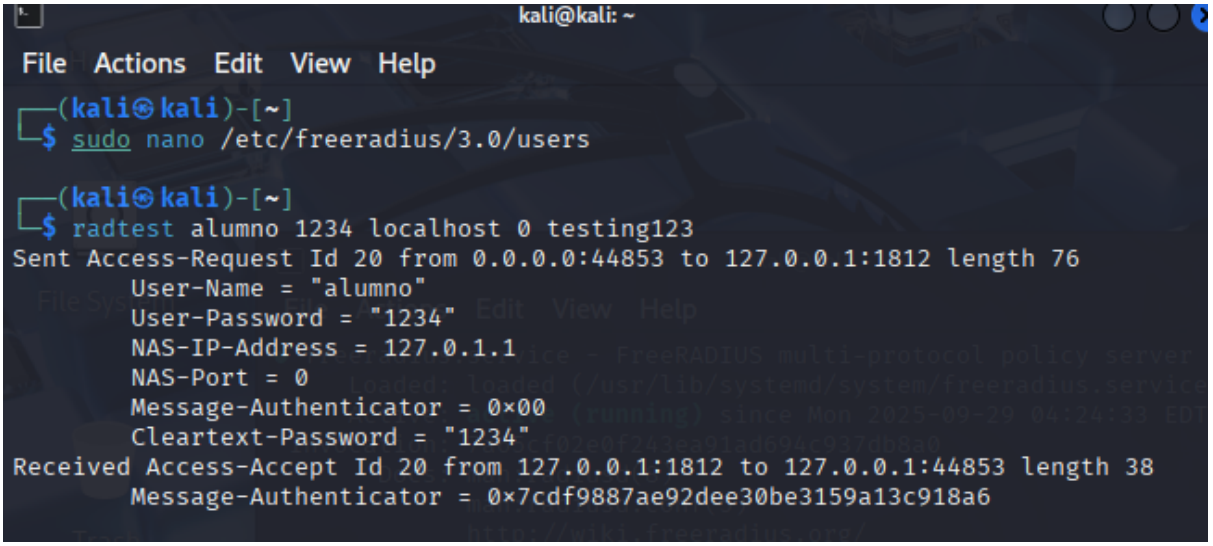


👉 Resultado esperado: usuario alumno con contraseña 1234 listo para autenticarse.

🔹 **Ejercicio 5 — Probar autenticación con radtest**

radtest alumno 1234 localhost 0 testing123

👉 Resultado esperado: respuesta Access-Accept.

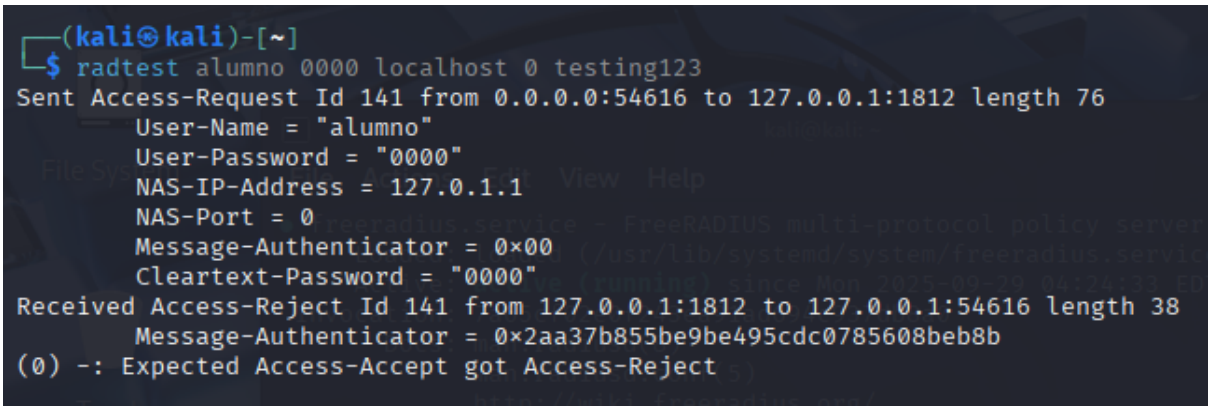


Esto es el caso de que haya sido incorrecto

🔹 **Ejercicio 6 — Probar con contraseña incorrecta**

radtest alumno 0000 localhost 0 testing123

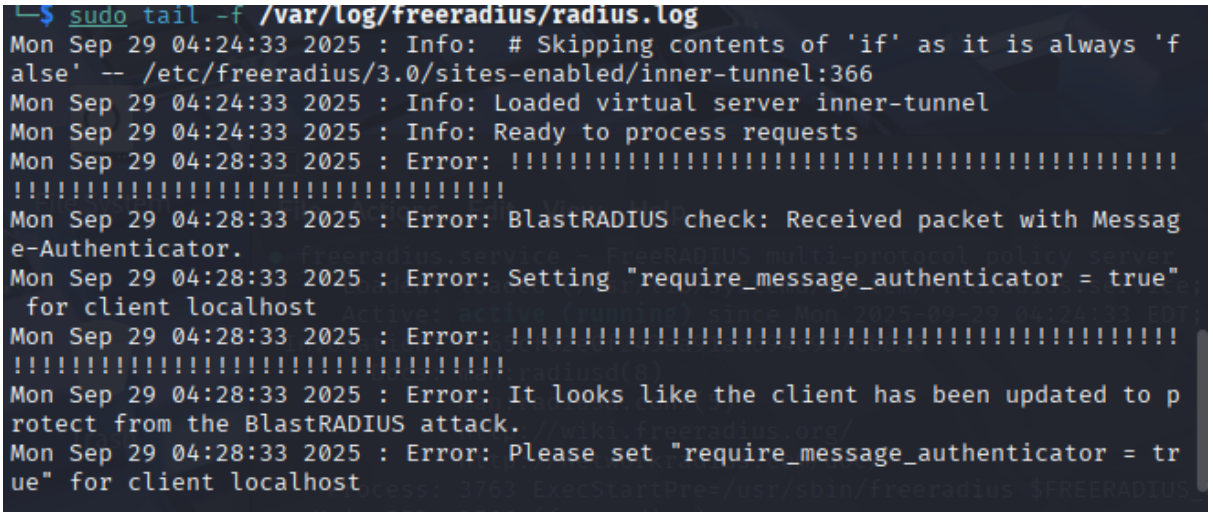
👉 Resultado esperado: respuesta Access-Reject.



🔹 **Ejercicio 7 — Revisar logs de autenticación**

sudo tail -f /var/log/freeradius/radius.log

👉 Resultado esperado: líneas mostrando quién intenta conectarse y con qué resultado.


$ sudo tail -f /var/log/freeradius/radius.log
Mon Sep 29 04:24:33 2025 : Info: # Skipping contents of 'if' as it is always 'f
alse' -- /etc/freeradius/3.0/sites-enabled/inner-tunnel:366
Mon Sep 29 04:24:33 2025 : Info: Loaded virtual server inner-tunnel
Mon Sep 29 04:24:33 2025 : Info: Ready to process requests
Mon Sep 29 04:28:33 2025 : Error: !!!!

```
┌──(kali㉿kali)-[~]
└─$ sudo systemctl restart freeradius

┌──(kali㉿kali)-[~]
└─$ radtest profesor abcd localhost:18120 0 testing123
Sent Access-Request Id 70 from 0.0.0.0:46945 to 127.0.0.1:18120 length 78
        User-Name = "profesor"
        User-Password = "abcd"
        NAS-IP-Address = 127.0.1.1
        NAS-Port = 0
        Message-Authenticator = 0x00
        Cleartext-Password = "abcd"
Received Access-Accept Id 70 from 127.0.0.1:18120 to 127.0.0.1:46945 length 38
        Message-Authenticator = 0x80a940a6e0952b53a4b33089c86bf483
```

🔹 **Ejercicio 9 — Cambiar puerto de escucha (opcional)**

1. Edita:

   sudo nano /etc/freeradius/3.0/sites-enabled/default

2. Busca listen { type = auth port = 1812 } y cámbialo a otro puerto (ej. 18120).
3. Reinicia y prueba con radtest:

   radtest alumno 1234 localhost:18120 0 testing123

```
┌──(kali㉿kali)-[~]
└─$ radtest alumno 1234 localhost:18120 0 testing123

Sent Access-Request Id 216 from 0.0.0.0:44139 to 127.0.0.1:18120 l
ength 76
        User-Name = "alumno"
        User-Password = "1234"
        NAS-IP-Address = 127.0.1.1
        NAS-Port = 0
        Message-Authenticator = 0x00
        Cleartext-Password = "1234"
Received Access-Accept Id 216 from 127.0.0.1:18120 to 127.0.0.1:44
139 length 38
        Message-Authenticator = 0x0341cbb13582a5594c55e042ff829a1d
```

🔹 **Ejercicio 10 — Detener y arrancar el servicio**

sudo systemctl stop freeradius

sudo systemctl start freeradius

👉 Resultado esperado: los alumnos comprueban cómo controlar el servicio.

```
┌──(kali㉿kali)-[~]
└─$ sudo systemctl stop freeradius

┌──(kali㉿kali)-[~]
└─$ sudo systemctl start freeradius
Job for freeradius.service failed because the control process exit
ed with error code.
See "systemctl status freeradius.service" and "journalctl -xeu fre
eradius.service" for details.
```

```
File  Actions  Edit  View  Help
└─$ sudo systemctl status freeradius
● freeradius.service - FreeRADIUS multi-protocol policy server
     Loaded: loaded (/usr/lib/systemd/system/freeradius.service; enabled; pres>
     Active: activating (auto-restart) (Result: exit-code) since Mon 2025-09-2>
 Invocation: 3316766b900e413d8ca54962bdcc4bc6
       Docs: man:radiusd(8)
             man:radiusd.conf(5)
             http://wiki.freeradius.org/
             http://networkradius.com/doc/
    Process: 17249 ExecStartPre=/usr/sbin/freeradius $FREERADIUS_OPTIONS -Cx ->
    Process: 17252 ExecStart=/usr/sbin/freeradius -f $FREERADIUS_OPTIONS (code>
   Main PID: 17252 (code=exited, status=1/FAILURE)
   Mem peak: 42.8M
        CPU: 244ms

Sep 29 05:58:28 kali systemd[1]: Failed to start freeradius.service - FreeRADI>
lines 1-15/15 (END)
```